

Math137A

Final Exam - June 12th, 2025

Instructor: Dr. Casteels Time: 180 minutes

Name:	
PERM:	

1. There are 44 points in total.
2. Clearly explain your answer. Write neatly!
3. Don't cheat.
4. If you are not sure what to do, partial points may be awarded by providing correct and relevant definitions or theorems.

1. (2 points each) **Each statement below is false.** For each statement, draw a **simple graph** that serves as a counterexample.

(a) A graph with n vertices and $n - 1$ edges is a tree.

(b) Every graph with an Eulerian circuit is bipartite.

(c) Every graph G satisfies $\chi(G) \leq \Delta(G)$.

(d) If G has n vertices and contains a Hamilton cycle, then every vertex $v \in V(G)$ has $\deg(v) \geq \frac{n}{2}$.

(e) If G has no perfect matching, then G must have an odd number of vertices.

(f) If G is 2-connected but not 3-connected, then G must have a vertex of degree 2.

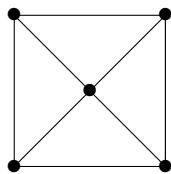
2. (4 points) Let G be a tree which has only vertices of degree 5 and degree 1. If G has 102 vertices, how many vertices of degree 5 are there and how many leaves are there?

3. (4 points) Let G be a connected planar graph with e edges, v vertices and where all faces have length at least 5. Prove that

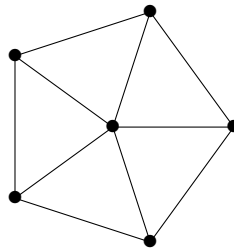
$$e \leq \frac{5}{3}(v - 2).$$

4. (4 points) A *wheel* graph W_n is constructed by taking an n -cycle, adding a new vertex and joining this vertex to every vertex. For example, W_4 and W_5 are below.

Calculate the chromatic number of W_n **for all** n . *Hint: the answer depends on whether n is even or odd.*

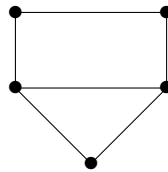


W_4



W_5

5. (4 points) How many ways are there to properly color the vertices of the following graph with a set of 8 colors? *You do not need to simplify your answer.*



6. (4 points) Suppose we have a set X of students and a set P of projects. Each student has exactly four projects they want to work on, and each project has **at most** four students who would be willing to work on that project.

Prove that the students can each be assigned a different project that they want to work on.

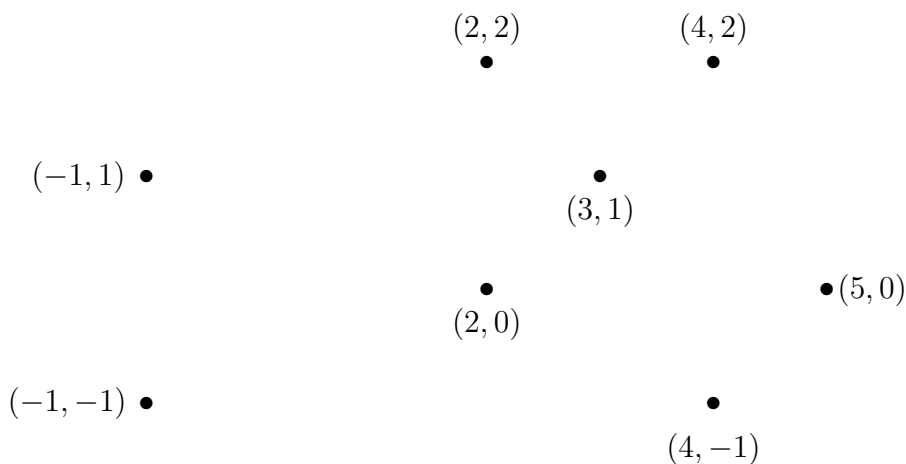
7. (2+2+2 points) As you should know well by now, the *Chebyshev Metric* on $\mathbb{Z}^2$ measures the distance between points (i, j) and (k, ℓ) to be

$$\max(|i - k|, |j - \ell|).$$

For example, the distance between $(1, 1)$ and $(2, 3)$ is 2. Given the points below, suppose all edges exist but are not drawn (in other words we have the complete graph) and that the cost of traveling between two points is given by the Chebyshev metric¹.

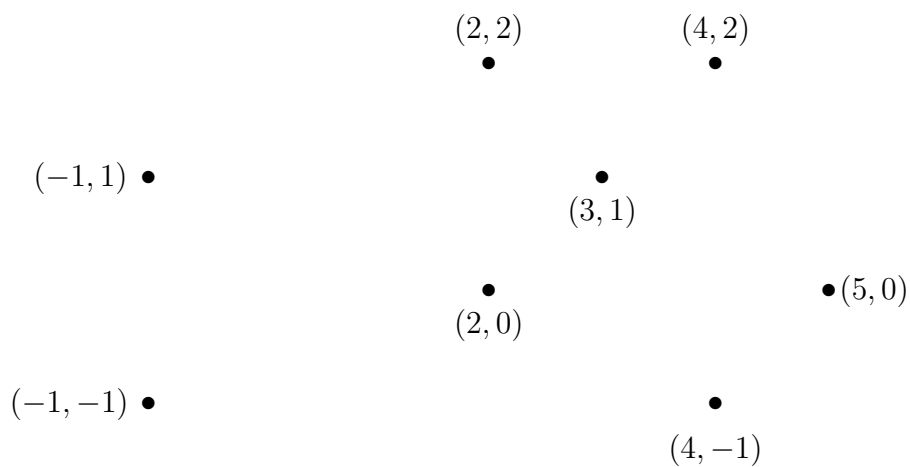
In this question use **Christofides' Algorithm** to find a Hamilton cycle whose cost is at most $3/2$ the cost of the cheapest Hamilton cycle.

- (a) Find a **minimum cost spanning tree** of the points below.

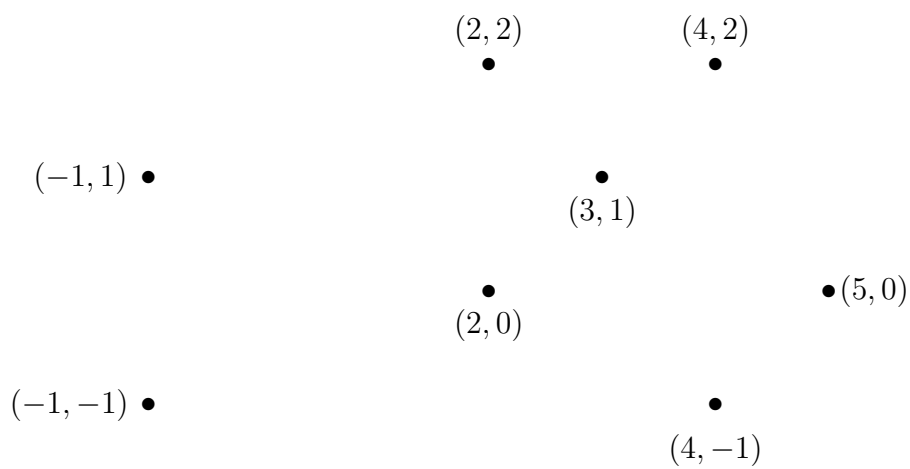


¹So you may assume the costs satisfy the triangle inequality.

- (b) **Copy the tree you found in Part (a) above**, and complete the next step of the algorithm here.

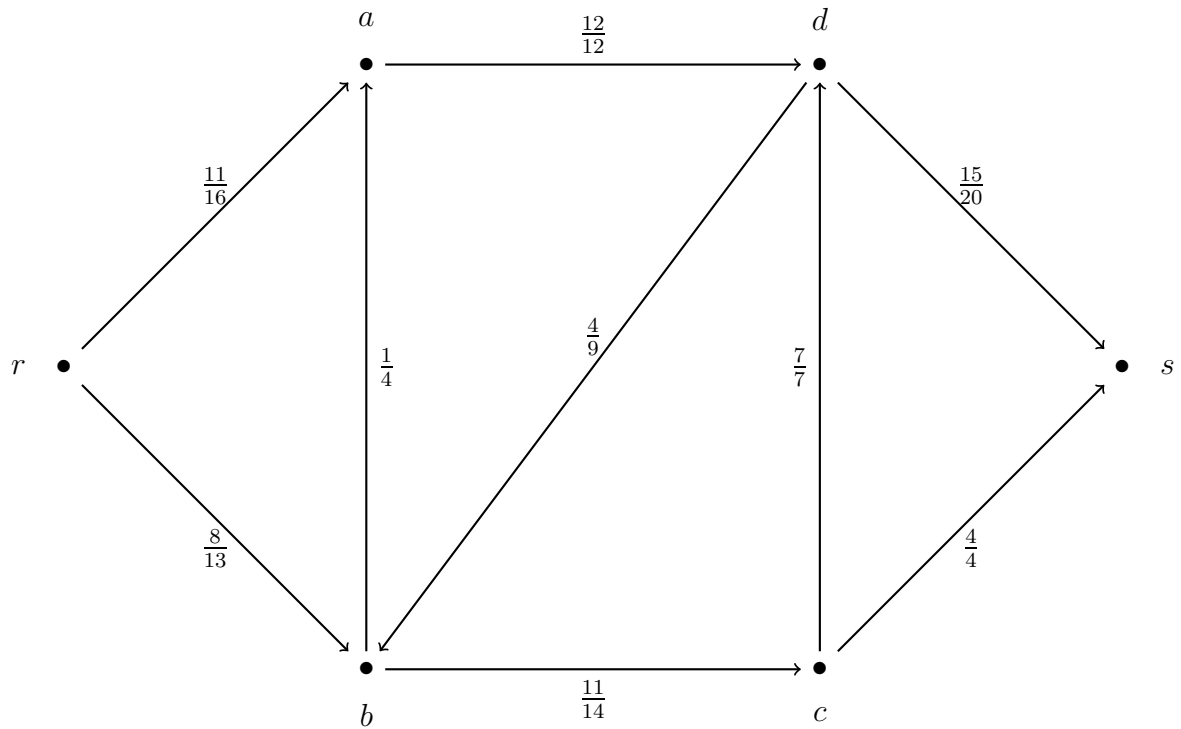


- (c) Copy your answer from part (b) here, and then complete the final step of the algorithm. **Indicate your starting point and direction of travel.**



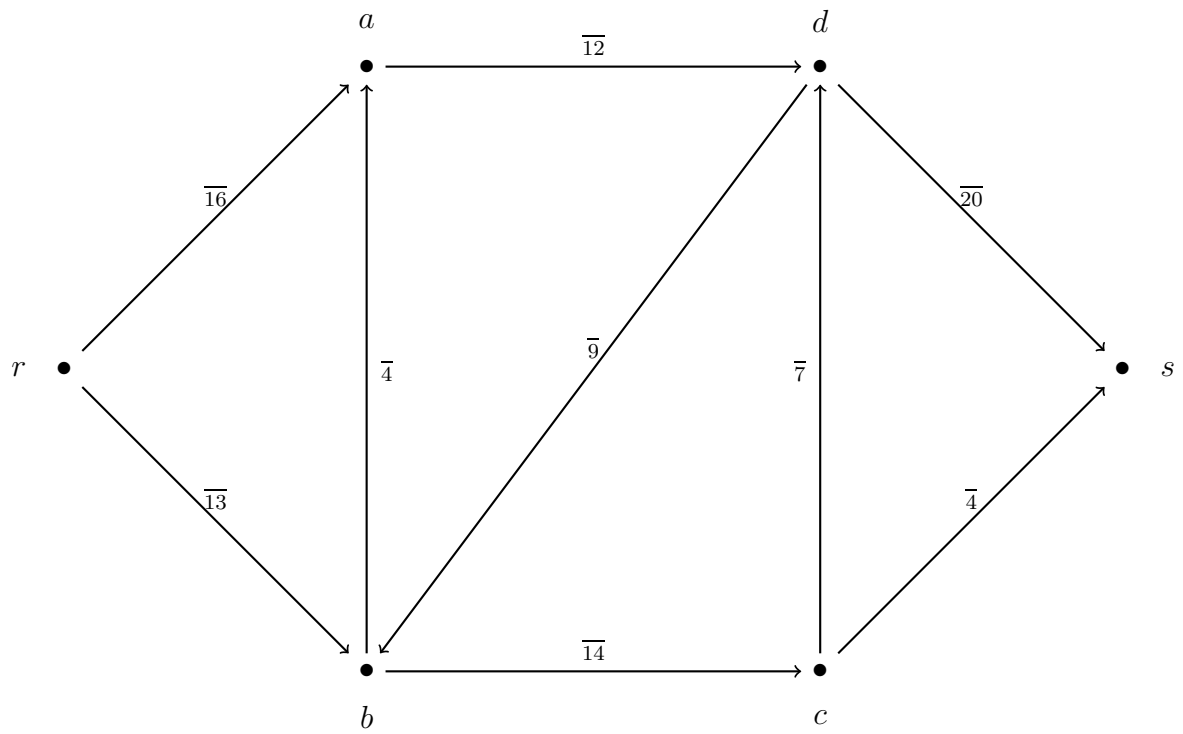
8. (2+2+2 points)

The following graph illustrates a flow from source r to vertex s . As usual, the numerators are the flow values, and the denominators are the capacities of the edges.



- (a) Explain why the current flow is *not* a maximum flow by finding an augmenting path from r to s .

- (b) Find a maximum (r, s) -flow. For convenience here is the same network again with only capacities shown.



- (c) Prove your flow is maximum by identifying a corresponding minimum (r, s) -cut (you can just circle the vertices in the above picture).